- **Biology**: The scientific study of life.
- **Systems Biology**: Construct models for the dynamic behavior of who biological systems.
- **Reductionism**: the reduction of complex systems to simpler components that are more manageable to study.
- **Eukaryotic cell** is subdivided by internal membranes.
- **Prokaryotic cell:** DNA is not separated from the rest of the cell in a nucleus.
- **Inquiry**: a search for information and explanation
- **Data**: Recorded observations
- **Hypothesis**: a tentative answer to a well-framed question.
- **Theory**: Much broader in scope than a hypothesis, is general enough to spin off many new, specific hypotheses, generally supported by a much greater body of evidence.
- **Technology**:applied scientific knowledge for some specific purpose

- **Evolution**: the process of change that has transformed life on Earth from its earliest beginnings to the diversity of organisms living today.
- **Properties of life**: Order, Evolutionary Adaption, Response to the Environment, Regulation, Energy processing, Growth and Development, Reproduction.

1. **<u>Themes connect the concepts of Biology</u>**
   1. New properties emerge at each level in the biological hierarchy (emergent properties). Thus, the study of life can be divided into different levels of biological organisation.
   2. Organisms interact with their environments, exchanging matter and energy: cycling of nutrients, flow of energy. Work requires a source of energy
   3. Structure and function are correlated at all levels of biological organization.
   4. Cells are an organism's basic units of structure and function.
   5. The continuity of life is based on heritable information in the form of DNA
      1. **Genes:** the units of inheritance that transmit information from parents to offspring.
      2. **DNA:** the substance genes are made of.
      3. **Genome**: The entire "library" of genetic instructions that an organism inherits.
   6. Feedback mechanism regulate biological systems.
      1. **Negative feedback**: Accumulation of an end product of a process slows that process.
      2. **Positive feedback**: Accumulation of an end product of a process speeds up that process.
2. <u>*The Core Theme:* **Evolution accounts for the unity and diversity of life**</u>
   1. **Organizing the diversity of life**
      1. *The Three Domains of Life:* Domain **Bacteria** and domain **Archaea** are prokaryotic. Domain **Eukarya** is eukaryotic.
      2. Domain **Eukarya** is further divided into the kingdoms *Protista, Plantea, Fungi,* and *Animalia.*
   2. **Charles Darwin and the Theory of Natural Selection**.
      1. Nov. 1859: Charles Robert Darwin publishes *On the Origin of Species by Means of Natural Selection.*
      2. "Decent with modification" and "natural selection"
   3. **The Tree of Life**
      1. Biologists' diagrams of evolutionary relationships generally take treelike forms.
3. **<u>Scientists use two main forms of inquiry in their study of nature</u>**
   1. **Discovery science** (descriptive science): Describing natural structures and processes as accurately as possible through careful observation and analysis of data.
      1. **Inductive reasoning**: Deriving generations from a large number of specific examples.
   2. **Hypothesis-based science**: Explaining nature.
      1. **Deductive reasoning**: From general premises, we extrapolate to the specific results we should expect if the premises are true.
      2. Hypotheses must be testable and falsifiable.
      3. Scientific method: The way scientists conduct experiments.
         1. **Controlled experiment**: Compares an experimental group with an control group.
4. **Scientific Models**: A tool used to help explain a scientific concept.

- **atom**: the smallest unit of matter that still retains the properties of an element.
- **Dalton**: a unit of measurement the same as the amu.
- **Energy:** the capacity to cause change.
- **Potential energy**: the energy that matter possesses because of its location or structure.
- **Chemical bonds**: Interactions that hold atoms together.
- **Valence**: the number of unpaired electrons required to complete an atom's outermost shell.
- **Electronegativity**: The attraction of a particular kind of atom for the electrons of a covalent bond.

## Chapter II: The Chemical Context of Life

*Living organisms are subject to basic laws of physics and chemistry.*

1. **Matter consists of chemical elements in pure form and in combinations called compounds**
   1. **Matter**: anything that takes up space and has mass.
   2. **Element**: a substance that cannot be broken down to other substances by chemical reactions.
   3. **Compound**: a substance consisting of two or more different elements combined in a fixed ratio.
   4. About 25% of the 92 elements are essential to life.
   5. Carbon, Hydrogen, Oxygen and Nitrogen make up 96% of living matter.
   6. Most of the remaining 4% is made up of phosphorous, potassium, and sulfer.
   7. **Trace elements:** elements required by an organism in only minute quantities.
2. **An element's properties depend on the structure of its atoms**
   1. **Subatomic particles:** neutrons (N), protons (+), electrons (-). Protons and neutrons are tightly packed together in the **atomic nucleus**.
   2. **Atomic number**: the number of protons in the nuclei of an atom.
   3. **Mass number**: the sum of the protons plus neutrons in the nucleus of an atom.
   4. **Atomic mass**: the mass of an atom (in daltons)
   5. **Isotope**: the different atomic forms of an element. A **radioactive isotope** is one in which the nucleus decays spontaneously.
   6. **Electron shells**: where electrons are found.
      1. **Valence electrons** are the outermost electrons and they are found in the **valence shell**.
      2. **Orbital:** the three-dimensional space where an electron is found 90% of the time. The first shell only has 1 *s* orbital. The second shell has 1 *s* orbital and 3 *p* orbitals. The third shell has more complex orbitals.
3. **The formation and function of molecules depend on chemical bonding between atoms**
   1. **Covalent bond**: the sharing of a pair of valence electrons by two atoms.
      1. **Molecule**: two or more atoms held together by covalent bonds.

- **Ion**: a charged atom or molecule
- **Cation**: Positively charged ion
- **Anion**: Negatively charged ion

- **Chemical reactions**: the making and breaking of chemical bonds.
- **Reactants**: starting materials
- **Products**: ending materials.
- **Chemical equilibrium**: the point at which the reactions offset one another equally

2. **Single bond**: only one pair of electrons shared.
3. **Structural Formula**: A way of representing molecular structure. Ex... H—H
4. **Molecular Formula**: Indicates what atoms are in a molecule. Ex.... $H_2$
5. **Double bond**: two pairs of electrons shared.
6. **Nonpolar covalent bond**: Electrons are shared equally
7. **Polar covalent bond**: Electrons are shared equally.

2. **Ionic Bonds**: In some cases, two atoms are so unequal in their attraction for valence electrons that the more electronegative atom strips an electron completely away from its partner. An **ionic bond** is formed when cations and anions attract each other.
   1. Compounds formed by ionic bonds are called **ionic compounds** or **salts**.
   2. The formula for an ionic compound indications only the ratio of elements in a crystal of the salt.
3. **Weak Chemical Bonds**
   1. **Hydrogen Bond**: Formed when a hydrogen atom covalently bonded to one electronegative atom is also attracted to another electronegative atom. In living cells, the electronegtaive atom is usually nitrogen or oxygen.
   2. **van der Waals interactions**: When atoms, by chance, become slightly polarized and molecules stick together slightly. Very weak force, but allows geckos to walk up a wall.
4. **Molecular Shape and Function**
   1. A molecule has a characteristic size and shape, determined by the positions of the atoms' orbitals. The shape of the molecule is very important to its function. Molecules with similar shapes have similar biological effects.
   2. For water ($H_20$), the molecule is shaped like a V with its two covalent bonds spread apart at an angle of 104.5º
   3. The methane molecule ($CH_4$) has the shape of a completed tetrahedron because all four hybrid orbitals of carbon are shared with hydrogen atoms.
   4. Only molecules with complementary shapes can form weak bonds with each other.

4. **Chemical Reactions Make and Break Chemical Bonds**
   1. All atoms of the reactants must be accounted for in the products: Reactions cannot create or destroy matter but can only rearrange it.
   2. All chemical reactions are reversible, with the products of the forward reaction becoming the reactants for the reverse reaction.

# Chapter 3: Water and the Fitness of the Environment

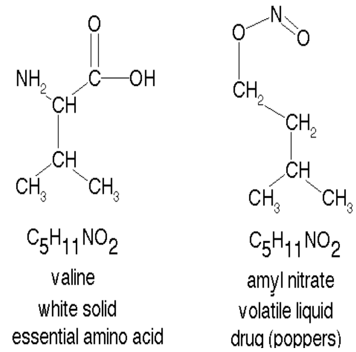
- Cells are about 70-95% water
- **Polar Molecule**: A molecule in which the two ends of the molecule have opposite charges.
- **Adhesion**: The clinging of one substance to another
- **Surface tension**: a measure of how difficult it is to stretch or break the surface of a liquid.
- **Kinetic energy**: the energy of motion.
- **Heat**: a measure of the matter's total kinetic energy due to motion of its molecules. Depends on the volume.
- **Temperature**: a measure of heat intensity that represents the average kinetic energy of the molecules, regardless of volume.
- **Heat of vaporization**: the quantity of heat a liquid must absorb for 1 g of it to be converted from the liquid to the gaseous state.
- **Solution**: a liquid that is a completely homogeneous mixture of two ore more substances.
- **Aqueous solution**: one in which water is the solvent.
- **Solvent**: the dissolving agent.
- **Solute**: the substance that is dissolved.
- Acid: a substance that increases the hydrogen ion concentration of a solution.
- Base: a substance that reduces the hydrogen ion concentration of a solution.
- **pH**: the negative common logarithm of the hydrogen ion concentration: pH=-log[$H^+$]. The lower the number, the more acid the solution. Most biological fluids are within the range pH 6-8.
- **Acid precipitation** refers to rain, snow or fog with a pH lower than pH 5.2. It affects lakes, streams and soil adversely.

2. **The Polarity of Water Molecules Results in Hydrogen Bonding**
   1. Water is a polar molecule. The slightly positive hydrogen of one molecule is attracted to the slightly negative oxygen of a nearby molecule.
3. **Four emergent properties of water contribute to Earth's fitness for life**
   1. *Cohesion:* a phenomenon that appears because hydrogen bonds hold water together.
      1. Cohesion due to hydrogen bonding contributes to the transport of water and dissolved nutrients against gravity in plants.
      2. Water also has a greater surface tension than most other liquids.
   2. *Moderation of Temperature:* Water moderates air temperature by absorbing heat from air that is warming and releasing the stored heat to air that is cooler. Water is good for heat storage because it has a high specific heat.
      1. **Calorie**: the amount of heat it takes to raise the temperature of 1 g of water by 1ºC.
      2. **Joule:** The work exerted when one applies a force of one newton to move an object 1m. 1 calorie equals 4.184 joules.
      3. **Specific Heat**: the amount of heat that must be absorbed or lost for 1g of a substance to change its temperature by 1ºC.
         1. Water's high specific heat is due to the hydrogen bonding that requires a lot more energy to break. This can moderate temperature near and in large bodies of water.
      4. **Evaporative Cooling:** as a liquid evaporates, the surface of the liquid that remains behind cools down.
         1. Water's high heat of vaporization is another emergent property caused by hydrogen bonds, which must be broken before the molecules can make their exodus from the liquid.
         2. Evaporative cooling of water contributes to the stability of temperature in lakes and ponds and also provides a mechanism that prevents terrestrial organisms from overheating.
   3. *Insulation of Bodies of Water by Floating Ice*
      1. Water is one of the few substances that are less dense as a solid than as a liquid. This is because hydrogen bonding locks solid water into a crystalline lattice and holds the molecules slightly apart. Water actually reaches its greatest density at 4ºC. Ice at 0ºC is actually 10% less dense than liquid water at 4ºC.
      2. Because ice floats on top of water, it insulates the water below it and prevents deep bodies of water from freezing completely, allowing life to survive under the frozen surface.
   4. *The Solvent of Life.*
      1. Water is the best solvent out there, but it is not a universal solvent.
      2. **Hydration shell**: the sphere of water molecules around each dissolved ion.
      3. **Hydrophilic**: Any substance that has an affinity for water. Substances can be hydrophilic without actually dissolving if the molecules are large enough.
      4. **Colloid**: a stable suspension of fine particles in a liquid.
      5. **Hydrophobic**: Substances that seem to repel water.
      6. **Molecular Mass**: the sum of the masses of all the atoms in a molecule.
      7. **Mole (mol)**: represents an exact number of objects—$6.022 \times 10^{23}$.
      8. **Molarity**: the number of moles of solute per liter of solution.
4. **Acidic and basic conditions affect living organisms**
   1. Occasionally, a hydrogen atom participating in a hydrogen bond between two water molecules shifts from one molecule to the other. The hydrogen leaves its electron behind.
      1. Hydroxide ion: $OH^-$ ; Hydronium ion:$H_3O^+$. By convention, it's represented by $H^+$, even though $H^+$ does not exist on its own in an aqueous solution. $OH^-$ and $H^+$ are very reactive.
      2. This is a reversible reaction. In pure water, only one water molecule in every 554 million is dissociated.
      3. Biologists use the pH scale to describe how acidic or basic a solution is.
   2. Strong acids and bases completely dissociate in water. Ex. HCl and NaOH. Weak acids and bases reversibly release and accept back hydrogen ions. Ex. $NH_4$ and $H_2CO_3$.
   3. In any aqueous solution at 25ºC, the *product* of the $H^+$ and $OH^-$ concentrations is constant at $10^{-14}$.
   4. The internal pH of most living cells is close to 7. The pH of human blood is very close to 7.4. A person cannot survive for more than a few minutes if the blood pH drops to 7 or rises to 7.8. **Buffers** are substances that minimize changes in the concentrations of $H^+$ and $OH^-$ in a solution.
   5. Considering the dependence of all life on water, contamination of rivers, lakes, seas and rain is a dire environmental problem.
   6. Carbon dioxide, the main product of fossil fuel combustion, also causes other problems. It is one of the chemicals implicated in global warming.
   7. When $CO_2$ dissolves in seawater, it reacts with water ($H_2O$) to form carbonic acid ($H_2CO_3$). Almost all of the carbonic acid in turn dissociates, producing protons and a balance between two ions, bicarbonate ($HCO_3^-$) and carbonate ($CO_3^{2-}$). As seawater acidifies due to the extra protons, the balance shifts towards $HCO_3^-$, lowering the concentration of $CO_3^{2-}$. Because calcification is directly affected by the concentration of $CO_3^{2-}$, any decrease in $CO_3^{2-}$ is therefore of great concern because calcification accounts for the formation of coral reefs in our tropical seas. The expected doubling of $CO_2$ emissions by the year 2065 could lead to a 40% decrease in coral reef calcification, according to a study by Chris Langdon and colleagues.

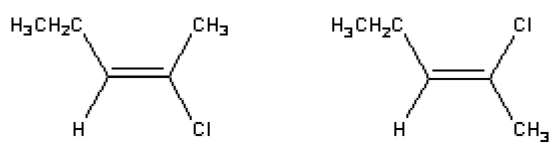
## Chapter 4: Carbon and the Molecular Diversity of Life

- **Organic chemistry**: the study of compounds containing carbon. Of all chemical elements, carbon is unparalleled in its ability to form molecules that are large, complex, and diverse.
- **Vitalism**: the belief in a life force outside the jurisdiction of physical and chemical laws
- **Mechanism**: the view that physical and chemical laws govern all natural phenomena.
- **Hydrocarbons**: organic molecules consisting of only carbon and hydrogen. They can release a relatively large amount of energy.

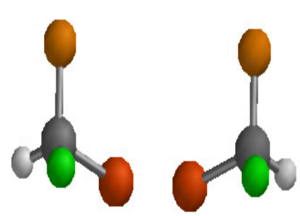
1. **Organic chemistry is the study of carbon compounds**
   1. The science of organic chemistry originated in attempts to purify and improve the yield of other organisms.
   2. Early 1800s: chemists could make many simple compounds but could not synthesize the complex molecules extracted from living matter.
   3. 1828: Friedrich Wöhler attempted to make ammonium cyanate by mixing solutions of ammonium ions ( $NH_4^{+¿}$ ) and cyanate ions ($CNO^-$). However, he managed to make urea instead.
   4. Hermann Kolbe, a student of Wöhler's made the organic compound acetic acid from inorganic substances that could be prepared directly from pure elements.
   5. 1953: Stanley Miller helped bring this abiotic synthesis of organic compounds into the context of evolution by simulating the early earth.
   6. All this discredited vitalism and supported mechanism.
2. **Carbon atoms can form diverse molecules by bonding to four other atoms**
   1. The key to an atom's chemical characteristics is its electron configuration. This configuration determines the kinds and number of bonds an atom will form with other atoms.
   2. Carbon usually completes its valence shell by sharing its 4 valence electrons. When a carbon atom forms four single covalent bonds, the bonds angle towards the corners of an imaginary tetrahedron. When two carbon atoms are joined by a double bound, however, all bonds around those carbons are in the same plane.
   3. Carbon chains form the skeletons of most organic molecules. The skeletons vary in length and may be straight, branched or arranged in closed rings.
   4. **Isomers**: compounds that have the same numbers of atoms of the same elements but different structures and hence different properties.
      1. **Structural isomers** differ in the covalent arrangements of their atoms.




      2. **Geometric isomers** have the same covalent partnerships, but they differ in their spatial arrangements.



      3. **Enantiomers** are isomers that are mirror images or each other.

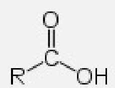
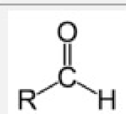
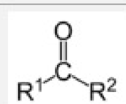
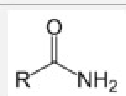
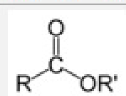


3. **A small number of chemical groups are key to the functioning of biological molecules**
   1. The distinctive properties of an organic molecule molecule also depend on the molecular components attached to the carbon skeleton.
   2. **Functional Group**: the chemical groups affect molecular function by being directly involved in chemicals reactions.

## Table of Functional groups

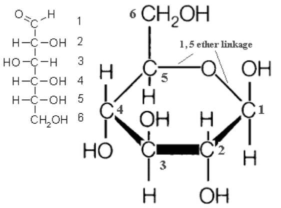
[edi

*Alk* is the prefix of the group (Meth, Eth, Prop, etc.)

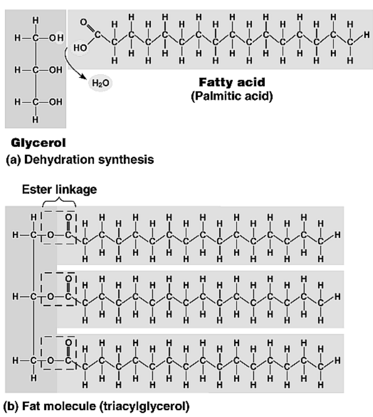
| Family | Structure | IUPAC nomenclature | IUPAC nomenclature for cyclic parent chains (if different from straight chains) | Common nomenclature |
|---|---|---|---|---|
| **Alkyl groups** | R— | ***Alk*yl** | - | ***Alk*yl** |
| **Halogens** | R—halogen | ***Halo'alk*ane** | - | ***Alk*yl *hal*ide** |
| **Alcohols** | R—OH | ***Alk*anol** | - | ***Alk*yl alcohol** |
| **Amines** | R—$NH_2$ | ***Alk*anamine** | - | ***Alk*yl amine** |
| **Carboxylic acids** |  | ***(Alk + 1)*anoic acid** | **Cyclo*alk*anecarboxylic acid** | - |
| **Aldehydes** |  | ***Alk*anal** | **Cyclo*alk*anecarboxaldehyde** | - |
| **Ketones** |  | ***Alk*anone** | - | ***Alk*$_{(1)}$yl *Alk*$_{(2)}$yl ketone** |
| **Thiols** | R—SH | ***Alk*anethiol** | - | - |
| **Amides** |  | ***(Alk + 1)*anamide** | **Cyclo*alk*anecarboxamide** | - |
| **Ethers** | $R_1$—O—$R_2$ | ***alk*oxy*alk*ane** | - | ***Alk*$_{(1)}$yl *Alk*$_{(2)}$yl ether** |
| **Esters** |  | ***Alk*$_{(1)}$yl *Alk*$_{(2)}$ aneoate** | ***Alk*$_{(1)}$yl Cyclo*alk*$_{(2)}$anecarboxylate** | ***Alk*$_{(1)}$yl *(Alk + 1)*$_{(2)}$ anoate** |

## Chapter 5: The Structure and Function of Large Biological Molecules.

- **Macromolecules**: enormous molecules that can have a mass well over 100,000 amu.




D-glucose



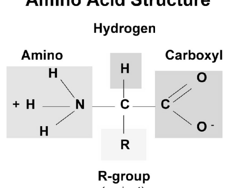

(a) Dehydration synthesis

(b) Fat molecule (triacylglycerol)



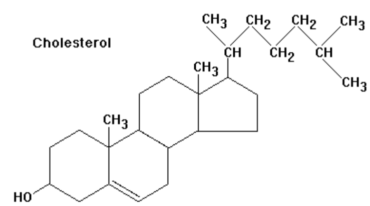
- **Enzymes**: biological **catalysts**. Most are proteins.
- **Amino acids**: organic molecules possessing both carboxyl and amino groups.

Amino Acid Structure



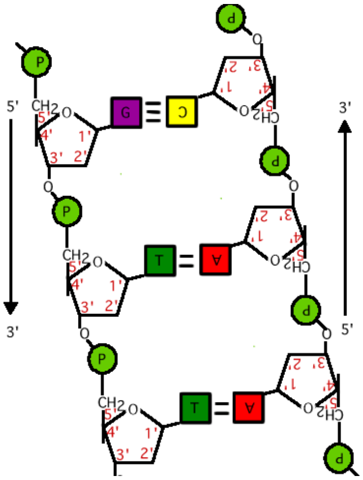

- **X-ray crystallography**: A method used to figure out the structure of molecules.
- **Nuclear Magnetic Resonance spectroscopy**: A method used to figure out the structure of molecules which does not require crystallization.
- **Bioinformatics**: Predicting the 3D structure of proteins from their amino acid sequences.

1. **Macromolecules are polymers, built from monomers**:
   1. **Polymer**: long molecule consisting of many similar or identical building blocks (**monomers**) linked by covalent bonds. Monomers are joined by **dehydration reactions**, which are facilitated by **enzymes**. The reverse reaction of dehydration synthesis is **hydrolysis**.
2. **Carbohydrates serve as fuel and building material**
   1. **Carbohydrates** include both sugars and polymers of sugars.
   2. The simplest carbohydrates are monosaccharides, which usually have empirical formulas of $CH_2O$. Glucose is the most common monosaccharide. It has a carbonyl group and multiple hydroxyl groups. Monosaccharides are classified by the number of carbon atoms and the location of the carbonyl groups.
      1. In aqueous solutions, most sugar molecules will form rings.
      2. Monosaccharides are major nutrients. Not only are they a major fuel for cellular work, they also serve as the raw material for the synthesis of other organic molecules.
   3. A **disaccharide** consists of two monosaccharides joined by a **glycosidic linkage**.
   4. **Polysaccharides** are macromolecules formed monosaccharides. They are generally used as energy storage and as structural materials.
      1. Plants tend to store **starch**, a polymer of glucose monomers. **Starch** is made of alpha-glucose and is helical. Generally, starch has few to no branching. The unbranched version is amylose; the branched version is amylopectin.
      2. Animals store **glycogen**, a more highly branched glucose polymer, instead.
      3. **Cellulose** is a major component of the tough walls that enclose plant cells. Because it is made of beta-glucose, the molecule is straight, and hydrogen-bonding between parallel cellulose molecules hold them together. About 80 cellulose molecules associate to form a microfibril, which weaves itself into the cell wall. Very few things can digest cellulose.
      4. **Chitin**: the carbohydrate used by arthropods to build their exoskeletons and fungi to build their cell ways.
3. **Lipids are a diverse group of hydrophobic molecules**
   1. **Fats** are constructed from glycerol, an alcohol with three carbons, each bearing a hydroxyl group; and **fatty acids**, long chains of carbon with a carboxyl group at one end. Fats separate water because the water molecules hydrogen-bond to each other and exclude the fats. Three fatty acids each join to a glycerol with ester linkages, forming a **triacylglycerol**.
      1. **Saturated fatty acid-**contains only single bonds between the carbon molecules. Usually solid at room temperature.
      2. **Unsaturated fatty acid**-contains one or more double bonds. A *cis* double bond creates a kink in the molecule, preventing the molecules from packing too tightly together so they are liquid at room temperature.
      3. Major function of fats is energy energy storage, insulation, and cushioning.
   2. **Phospholipids** make up the cell membranes. They have two fatty acids joined to a glycerol; the third hydroxyl group of glycerol is joined to a phosphate group. The fatty acids are hydrophobic, the phosphate group is hydrophilic. In water, phospholipids form a lipid bilayer to keep the fatty acid tails away from water.
   3. **Steroids**: lipds characterized by a carbon skeleton consisting of four fused rings.
      1. **Cholesterol** is a common component of animal cell membranes and is also the precursor from which other steroids are synthesized.




4. **Proteins have many structures, resulting in a wide range of functions**
   1. Proteins account for more than 50% of the dry mass of most cells. Some proteins speed up chemical reactions, while others play a role in structural support, storage, transport, cellular communication, movement, and defense against foreign substances.
   2. All proteins are polymers constructed from the same set of 20 amino acids. A protein consists of one or polypeptides. A functional protein is not just a polypeptide chain, but one or more polypeptides precisely twisted, folded, and coiled into a molecule of unique shape. There are four levels of protein structure:
      1. Primary structure: the unique sequence of amino acids.
      2. Secondary Structure: either a coil (alpha-helix) or folds (beta-pleated sheet) that result from the hydrogen bonds between the repeating constituents of the polypeptide backbone.
      3. Tertiary structure: the overal shape of a poly peptide resulting from interactions between the side chains (R groups).
         1. **Hydrophobic interaction:** water molecules exclude non-polar substances as they form hydrogen bonds with each other and the hydrophilic parts of the protein. Non-polar amino acids then attract each other with van der Waals intercations.
         2. **Polar side chains and ionic bonds:** form between positively and negatively charged side chains to help stabilize tertiary structure.
         3. **Disulfide bridges:** the sulfur of cysteine bonds to the sulfur of another cysteine.
      4. Quaternary structure: the overall protein strcture that results from the aggregation of several polypeptide subunits.
   3. Proteins are joined by dehydration synthesis, which forms **peptide bonds.**
   4. **Denaturation:** the processes that causes a protein to lose its structure and function.
   5. **Chaperonins:** proteins molecules that assist in the proper folding of other proteins.

- **Gene**: a unit of inheritance. Codes for the amino acid sequence of a polypeptide.




5. **Nucleic acids store and transmit hereditary information**
    1. **Nucleic acids**: enable living organisms to reproduce their complex components from one generation to the next. There are two types:
        1. **Deoxyribonucleic acid**
        2. **Ribonucleic acid**
    2. Sites of protein synthesis are tiny structures called ribosomes.
    3. Nucleic acids are macromolecules that exist as polymers called **polynucleotides,** which consist of monomers called **nucleotides**.
        1. There are two types of nucleotides:
            1. **pyrimidines**, which have a six-membered ring of carbon and nitrogen atoms. They include cytosine, thymine, and uracil.
            2. **Purines**: which have a six-membered ring fused to a five-membered ring. They include adenine and guanine.
        2. The sugars connected to the nitrogenous base are **ribose** and **deoxyribose;** the later lacks an oxygen atom on the second carbon of its ring. Because the atoms in both the nitrogenous base and the sugar are numbered, the sugar atoms have a prime (') after the number to distinguish them.
        3. Adjacent nucleotides are joined by a phosphodiester linkage. The two free ends of the polymer are distinctly different from each other. One end has a phosphate attached to a 5' carbon, and the other end has a hydroxyl group on the 3' carbon.
    4. DNA molecules have two polynucleotides that spiral around an imaginary axis, forming a double helix.
        1. The two sugar-phosphate backbones run in opposite 5'→3' directions from each other, an arrangement referred to as **antiparallel.**
        2. One long DNA double helix includes many genes.
        3. Adenine always pairs with thymine, and guanine always pairs with cytosine (in DNA).
    5. The linear sequences of nucleotides in DNA molecules are passed from parents to offspring. Siblings have greater similarity in their DNA and proteins than do unrelated individuals of the same species.
    6. Molecular biology has added a new tape measure to the toolkit biologists use to assess evolutionary kinships.